# Representing sound; buffering and streaming

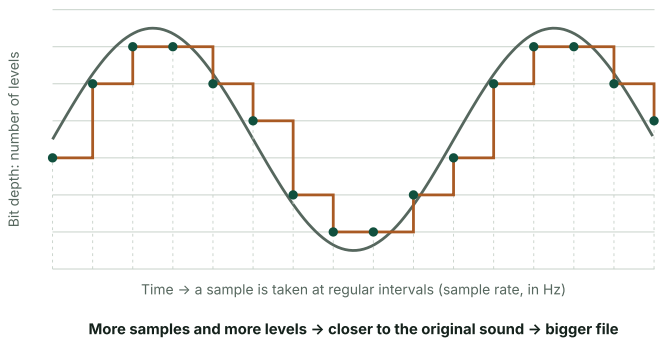
L04 FACT FILE

**Specification: students should be able to**

- Explain the need for analogue-to-digital conversion in sound recording
- Describe factors that affect sound quality when recording sound, including sample rate, bit depth and bit rate
- Calculate the bit rate and file size of a sound recording
- Demonstrate understanding of how buffering and streaming are used to support the transfer of moving image files

CCEA fact file: Digital Data 03: Representing Sound

**Key words:** analogue signal, digital, microphone, analogue-to-digital converter (ADC), sampling, sample, amplitude, sample rate, hertz (Hz), bit depth, bit rate, channels (mono/stereo), file size, streaming, downloading, buffer, buffering




A sound wave sampled at regular intervals, with sample rate on the time axis and bit depth as the number of levels

## How a computer records sound

Sound is a wave of vibrations. It is an analogue signal, which means it changes smoothly and continuously. A computer can only work with digital data, which is made of binary 0s and 1s, so the sound has to be converted before it can be stored. First a microphone picks up the sound and turns it into an analogue electrical signal. Then an analogue-to-digital converter (ADC) measures the height of the signal at regular, equal time intervals. Each measurement is called a sample. The ADC rounds each sample to the nearest whole-number level and stores that number in binary. The finished recording is just a long list of binary numbers. Because the wave is only measured at set moments, anything the wave does between two samples is not recorded, so a digital recording is always an approximation of the original sound. How good that approximation is depends on the sample rate and the bit depth.

## Sample rate, bit depth and bit rate

Sample rate is the number of samples taken each second. It is measured in hertz (Hz), so 44 100 Hz means 44 100 samples per second, which is the standard for CDs. More samples per second follow the shape of the wave more closely, so the sound is more like the original. Bit depth is how many bits are given to each individual sample. With 8 bits there are 256 possible levels, but with 16 bits there are 65 536, so each sample is rounded much less and is stored more accurately. Bit rate is the number of bits processed for every second of sound. For one channel it is the sample rate multiplied by the bit depth, and for stereo you multiply by two because there is a left and a right channel. Increasing any of these three improves the quality, but it also increases the file size, so the recording takes up more storage and takes longer to send.

### Working out file sizes

To find the size of an uncompressed sound file, multiply the sample rate by the bit depth by the number of channels by the length of the recording in seconds. The answer is in bits. Divide by 8 to change bits into bytes, then by 1024 to change bytes into kilobytes, and by 1024 again to get megabytes. For example, a 30-second mono clip recorded at 11 025 Hz with an 8-bit depth needs 11 025 × 8 × 30 = 2 646 000 bits. Dividing by 8 gives 330 750 bytes, which is about 323 KB. Always write the formula down first, then put in the numbers, and keep track of the units at each step. A very common mistake is forgetting to change minutes into seconds, so a 2-minute track must use 120 seconds. Another is forgetting to divide by 8. Compressed formats such as MP3 store much less than this, because some data is removed.

### Streaming and buffering

Video files are very large, so downloading a whole film before watching it can take a long time. Streaming solves this. The video is sent over the internet in small pieces and starts playing as soon as the first pieces arrive. The user watches in real time while the rest is still being received, and the file is not normally saved permanently on the device. To make streaming smooth, the device uses a buffer. This is an area of memory that stores the next part of the video before it is needed. While one section is playing, the following section is downloaded into the buffer. If the internet connection slows down or drops briefly, the video keeps playing from the buffer, so the viewer does not notice. Only if the buffer empties completely does the video stop and show a loading icon. Streaming needs a constant connection, which can be a problem in rural areas with weak broadband or mobile signal.

## T Key terms from the mark schemes

**Analogue signal**
A signal that varies smoothly and continuously, representing a physical quantity such as a sound wave.
`Credited: continuously varying signal`

**Analogue-to-digital converter (ADC)**
A device that converts an analogue signal, such as sound from a microphone, into digital (binary) data by sampling it at regular intervals.
`Credited: U1 2022 Q1, 2023 Q2`

**Sampling**
Measuring the level (amplitude) of an analogue signal at regular time intervals; each reading is rounded to a whole-number level and stored in binary.

**Sample rate**
The number of samples of a sound taken each second, measured in hertz (Hz). CD audio uses 44 100 Hz (44.1 kHz).
`Credited: U1 2023 Q2, 2025 Q2; sampling frequency`

**Bit depth**
How many bits are allocated to every sound sample. More bits give more possible levels, so each sample is recorded more accurately.
`Credited: U1 2024 Q7, 2025 Q2`

**Bit rate**
The number of bits of data processed for each second of sound, usually in kilobits per second; for one channel it equals sample rate × bit depth.
`Credited: U1 2019 Q1, 2022 Q1`

**Streaming**
Playing video or audio over the internet while it is still being received, without waiting to download and permanently store the whole file.
`Credited: U1 2019 Q4`

**Buffer / buffering**
An area of memory that temporarily stores the next part of a video or song before it is played, so playback continues if the connection slows or drops briefly.
`Credited: U1 2019 Q4, 2021 Q11, 2022 Q1, 2023 Q2`

**Stereo**
Sound recorded on two channels (left and right); it needs twice as much data as the same sound in mono (one channel).

### Exam focus

This lesson comes up almost every year. Bit rate has appeared as a multiple-choice question (2019 Q1(j), 2022 Q1). The purpose of an ADC was tested in 2022 Q1 and 2023 Q2(e): say it converts the analogue signal into digital (binary) data by sampling it at regular intervals. Sample rate was asked in 2023 Q2(d) and bit depth in 2024 Q7(b); for each, give a clear definition and then say how increasing it affects both quality and file size. In 2025 Q2(b)–(c) pupils had to calculate the size of a sound file, so show every step: the formula, the multiplication, then the conversion from bits to bytes and KB. Streaming and buffering were tested in 2019 Q4(b), 2022 Q1 and 2023 Q2(c). Marks come from saying that the buffer is temporary memory, that it stores the next part of the video in advance, and that playback continues from it if the connection slows. Do not mix up bit depth (bits per sample) with bit rate (bits per second).

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|---|---|---|
| Unit 1 2025 Q2 | [6] | Bitmap vs vector; sound file size calculation |
| Unit 1 2024 Q7 | [5] | Compression, video formats, bit depth |
| Unit 1 2023 Q2 | [10] | Bitmap vs vector, resolution, buffering, sample rate, ADC, compression |
| Unit 1 2022 Q1 | [9] | MCQ: storage, bit rate, real-time, compression, buffering, ADC, big data, backup |
| Unit 1 2021 Q11 | [8] | Key term matching |

## ? Check yourself

1. Explain why analogue-to-digital conversion is needed when recording sound on a computer. [2]
2. State the purpose of an analogue-to-digital converter (ADC). [1]
3. What is meant by sample rate? State the unit it is measured in. [2]

**Answers:** 1. Sound is an analogue signal / continuous wave (1) / computers can only store and process digital (binary) data, so the sound must be converted (1). 2. To convert an analogue signal (e.g. from a microphone) into digital / binary data (by sampling it) (1). 3. The number of samples taken per second (when recording sound) (1) / measured in hertz (Hz) (1).